

CBSE Class 10 Mathematics

Time Allowed: 3 Hours

Maximum Marks: 10

General Instructions:

1. This question paper contains 10 questions. All questions are compulsory.
2. Questions are grouped into sections A through E.
3. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. The quadratic equation $kx^2 - 6x + 1 = 0$ has real and equal roots. What is the value of k ?
(A) (A) 3
(B) (B) 6
(C) (C) 9
(D) (D) 36
2. If one root of the quadratic equation $x^2 - 8x + p = 0$ is 3, then the value of p is:
(A) (A) 5
(B) (B) 15
(C) (C) -5
(D) (D) -15
3. Which of the following is a quadratic equation?
(A) (A) $(x + 1)^2 = 2(x - 3)$
(B) (B) $x^2 + 3x = (x + 1)^2$
(C) (C) $x + \frac{1}{x} = x^2$
(D) (D) $(x + 2)^3 = x^3 + 1$
4. If the equation $x^2 + 2(k + 2)x + 9k = 0$ has equal roots, then k can be:
(A) (A) 1
(B) (B) 4
(C) (C) -1, -4
(D) (D) 1, 4
5. The discriminant of the quadratic equation $3x^2 - 2x + \frac{1}{3} = 0$ is:
(A) (A) 0

- (B) (B) 1
- (C) (C) 2
- (D) (D) -1

6. Which constant must be added to $x^2 + 6x$ to make it a perfect square?

- (A) (A) 3
- (B) (B) 9
- (C) (C) 6
- (D) (D) 36

7. If the sum of the roots of the equation $ax^2 + bx + c = 0$ is equal to the product of the roots, then:

- (A) (A) $b = c$
- (B) (B) $b = -c$
- (C) (C) $b + c = 0$
- (D) (D) $b - c = 0$

8. The quadratic equation $2x^2 - px + 8 = 0$ has equal roots. What is the value of p ?

- (A) (A) 8
- (B) (B) -8
- (C) (C) ± 8
- (D) (D) 0

9. If $x = 2$ is a root of $x^2 + kx + 4 = 0$, the value of k is:

- (A) (A) 4
- (B) (B) -4
- (C) (C) 2
- (D) (D) -2

10. The roots of the equation $x^2 - 3x = 0$ are:

- (A) (A) 0, 3
- (B) (B) 0, -3
- (C) (C) 3, 3
- (D) (D) 1, 3